TINY – OBJECT DECTION

Where is waldo

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LEFT: YOLO

RIGHT: AGENT

Code and results

Abstract

Finding Waldo is a challenging tiny-object detection task in highly crowded scenes. We compare two approaches: a tiled YOLO26s detector and a training-free vision-language agent.
YOLO26s was trained on tiled Waldo images. The best 416 × 416 centred-tile configuration achieved precision of 0.857, recall of 0.508, mAP@0.5 of 0.544 and mAP@0.5:0.95 of 0.228. The agent uses overlapping patches and Gemini-based reasoning to identify and verify Waldo candidates without task-specific training.
Both approaches show that tiling helps preserve small visual details for crowded visual search.

Motivation

Inspired by the classic childhood *Where's Waldo?* books and visual-search game, we wanted to explore whether AI could solve the same deceptively simple puzzle. Finding Waldo requires careful visual search: he is very small within a crowded scene, may be partly hidden by other characters, and shares familiar features—such as red-and-white stripes, glasses and a hat—with many distractors. These scenes turn a playful childhood challenge into an interesting AI problem involving both detailed visual reasoning and tiny-object localisation. We found this challenge both technically demanding and genuinely fun, which inspired this project.

Methodology

- **Method A — Tiled YOLO26s Detector:** We train YOLO26s as a single-class Waldo detector using object-centred tiles and carefully selected background-only tiles. We evaluate different settings if tile size and select the strongest validation configuration, then map tile-level detections back to their original image positions.
- **Method B — Gemini VLM Workflow:** Each full-resolution scene is deterministically split into overlapping 256 × 256 patches with 15% overlap. Gemini 3.5 Flash independently answers a simple binary question for each patch: *"Is Waldo present?"* The patches are evaluated concurrently to identify candidate locations efficiently.

Method A - Tiled TOLO26 detector

Train on object-centered tiles, then merge tile detections into the original scene

- **Tile preparation:** Each crowded scene is split into 416 × 416 tiles, including tiles containing Waldo and carefully selected background-only tiles. This keeps small details, such as Waldo's striped shirt and glasses, visible.
- **YOLO26s detection:** YOLO26s is trained to detect Waldo as a single object class, then examines each overlapping tile independently during testing.
- **Final localization:** Detections are placed back in their original positions. Duplicate overlapping boxes are removed, leaving one final bounding box around Waldo.

Method B - Training-free VLM agent

Ask a vision-language model which tile holds Waldo, then force one choice among the candidates

- **Tile preparation:** A fixed 256 × 256 sliding window sweeps the scene at 15% overlap, with the last row and column snapped to the edge so no region is missed. No training data and no labelled boxes are required.
- **VLM detection:** Each tile is sent to gemini-3.5-flash with one binary question — is Waldo here? Only the yes/no answer is used, as the model's own confidence score proved unreliable.
- **Cross-candidate verification:** When several tiles say yes, all candidates are re-cropped and compared in a single request, forcing one final choice; the winner is mapped back to original-image coordinates.

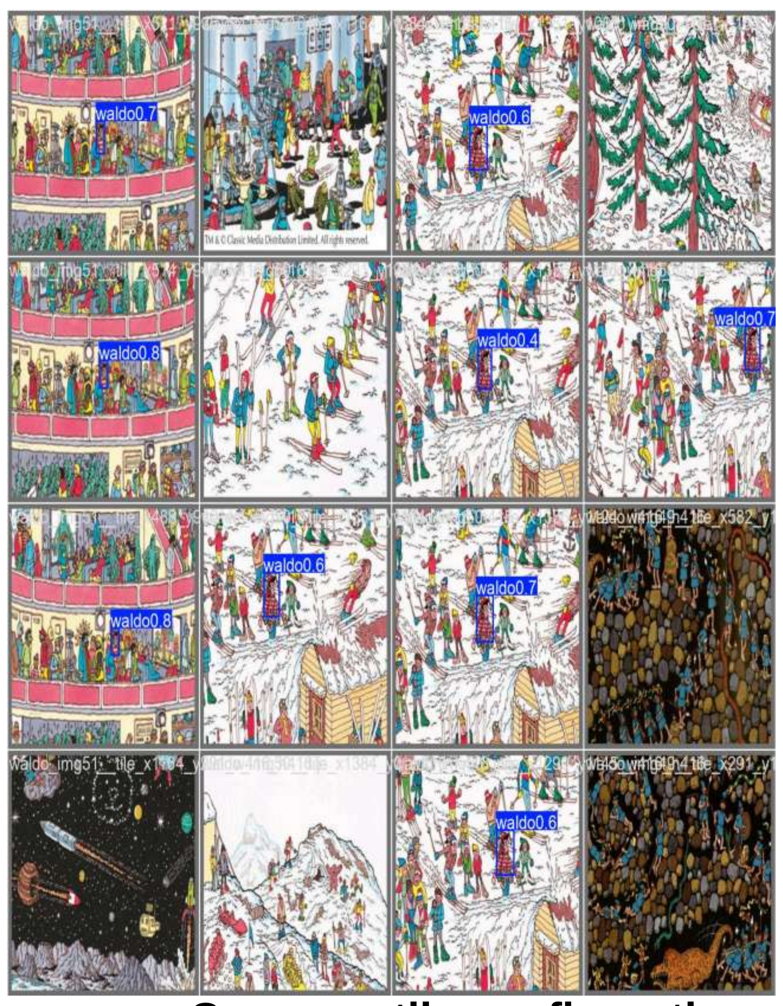
TRAINING-FREE VLM WORKFLOW

Deterministic control flow; Gemini is used only as a classifier/comparator

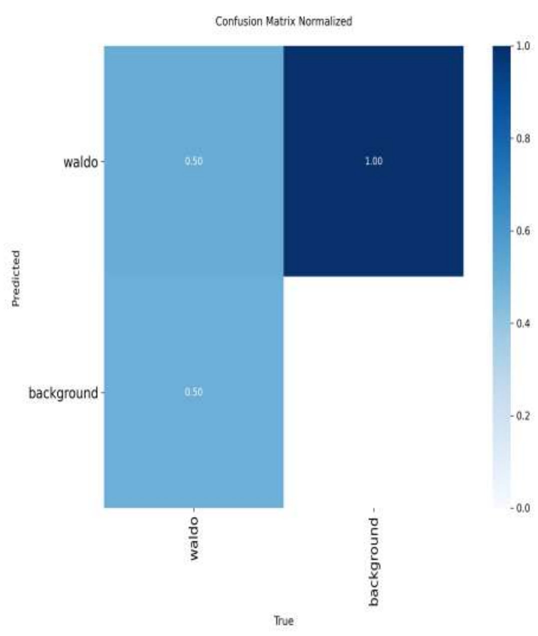
FULL SCENE	256 px TILES	VLM DETECT	VERIFY IF >1	ORIGINAL BBOX
Crowded high-resolution illustration	15% overlap; final row/column snapped to edge	Gemini binary present signal; 10 workers	Compare all candidates in one request	Map winner back and draw final box

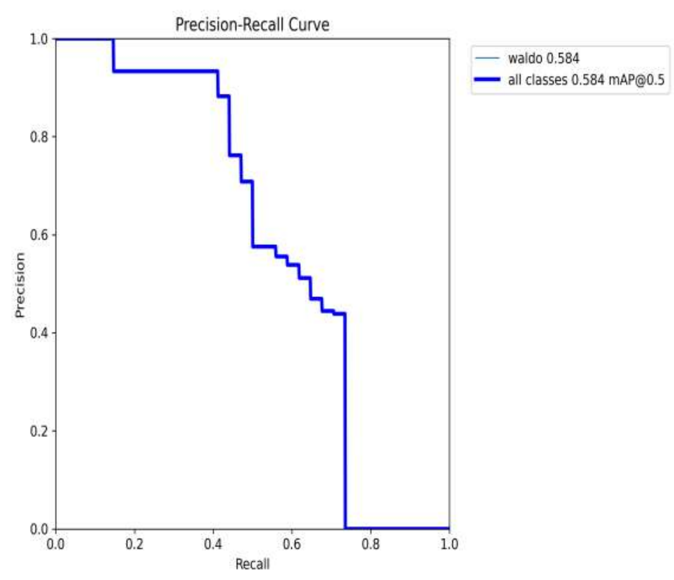
Fast path: 0 or 1 candidate skips verification

YOLO prediction result



Dataset	Best epoch	Precision	Recall	mAP50	mAP50–95
Data_tiled_768	65	0.49890	0.35644	0.28967	0.1577
Data_tiled_416_center_neg05	55	0.85744	0.50774	0.54352	0.2278





- **Compare tile configurations:** We trained YOLO26s using two tiled datasets: data_tiled_768 and data_tiled_416_center_neg05. The 768-pixel setting achieved mAP@0.5 of **0.290**, while the 416-centred setting achieved **0.544**.
- **Select the 416 × 416 configuration:** The data_tiled_416_center_neg05 dataset produced the best result at **epoch 55**. Compared with 768-pixel tiles, 416-centred tiles improved precision from **0.499** to **0.857** and recall from **0.356** to **0.508**.
- **Final selected-model performance:** The best 416-centred YOLO26s model achieved **Precision = 0.857**, **Recall = 0.508**, **mAP@0.5 = 0.544**, and **mAP@0.5:0.95 = 0.228**.

VLM WORKFLOW EXPERIMENT RESULTS

Three representative qualitative outputs

94.4%	4.9%
PATCH RECALL	FALSE-POSITIVE RATE
~33 s	~$0.09
FAST-PATH LATENCY	API COST / SCENE

- **Detection performance:** The VLM workflow reached 94.4% patch recall with a 4.9% false-positive rate.
- **Efficient fast path:** A single-candidate scene completes in about 33 s at an estimated API cost of $0.09..
- **Reliable decision signal:** Confidence contradicted the binary present flag in 77% of evaluated patches, so filtering uses only the binary answer.

Patch-level detection metrics; end-to-end bounding-box IoU has not yet been quantified.

Conclusion and Future work

Conclusion. This project implements two approaches for locating Waldo in crowded scenes: a tiled YOLO26s detector and a Gemini VLM workflow. The YOLO approach shows that tiling can preserve small visual details, but its main limitation is the small training dataset. Data augmentation was therefore essential to provide more varied training examples. The VLM workflow offers a training-free, patch-based approach that can identify and verify likely Waldo locations.
Future Work. We will quantitatively compare the two approaches using shared evaluation images and localisation metrics. We will also investigate additional augmentation strategies, such as colour variation, cropping, scaling, occlusion and synthetic scene generation, to improve the robustness and recall of the YOLO detector.

Reference

[1] B. H. Miao and A. Lesh, "Development of WaldoNet: A Novel Approach to Solving 'Where's Waldo?'" Stanford CS231N, 2024.
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Github Connect

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